

STAT-437: APPLIED STATISTICAL METHODS

Assignment: 2

Due: October 2, 2026 at 10:00PM

Total=40 Marks

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Instructions: All answers and relevant R outputs must be given in the main part. Include R code in the Appendix and submit the source code in the script(.r) or markdown(.rmd). Also, pdf version of Assignment is required to submit. Late submissions will not be accepted.

Question-1:

Suppose a Generalized Linear Model is fitted under the hypotheses

$$H_0 : \mathbf{R}\boldsymbol{\beta} = 0 \quad vs \quad H_1 : \mathbf{R}\boldsymbol{\beta} \neq 0$$

where $\mathbf{R}$ is a known $p_2 \times p$ matrix of rank p_2 , and the model under H_0 is a restricted model nested within the full model under H_1 .

Let $\hat{\boldsymbol{\beta}}$ denote the maximum likelihood estimator under the full model and $\tilde{\boldsymbol{\beta}}$ denote the maximum likelihood estimator under the restriction $\mathbf{R}\boldsymbol{\beta} = 0$. The deviance (likelihood-ratio statistic) is defined by

$$D = 2\{l(\hat{\boldsymbol{\beta}}, \phi) - l(\tilde{\boldsymbol{\beta}}, \phi)\}$$

where $l(\hat{\boldsymbol{\beta}}, \phi)$ and $l(\tilde{\boldsymbol{\beta}}, \phi)$ are the maximized log-likelihoods under the full and restricted models, respectively.

- (i) Explain why the deviance is always non-negative.
- (ii) Explain what a large value of the deviance indicates about the restricted model relative to the full model.
- (iii) Under H_0 , what is the approximate distribution of the deviance for testing p_2 restrictions?
- (iv) Explain how the deviance can be used to determine whether the predictors removed from the full model result in a statistically significant loss of fit.

Question-2:

Article production by graduate students in biochemistry Ph.D. programs:

"bioChemists" data set is available in the r-package "pscl". The description, Usage, Format and References of bioChemists dataset is as below:

- Description: A sample of 915 biochemistry graduate students.
- Usage: data(bioChemists)
- Format:
 - art: count of articles produced during last 3 years of Ph.D.
 - fem: factor indicating gender of student, with levels Men and Women
 - mar: factor indicating marital status of student, with levels Single and Married
 - kid5: number of children aged 5 or younger
 - phd: prestige of Ph.D. department
 - ment: count of articles produced by Ph.D. mentor during last 3 years

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- References:

(i) Long, J. Scott. 1990. The origins of sex differences in science. *Social Forces*. 68(3):1297-1316.

(ii) Long, J. Scott. 1997. *Regression Models for Categorical and Limited Dependent Variables*. Thousand Oaks, California: Sage.

Give the answers of the following:

(i) Consider the Poisson GLM

$$\log\{E(art_i)\} = \beta_0 + \beta_1 fem_i + \beta_2 mar_i + \beta_3 phd_i + \beta_4 ment_i, \quad i = 1, 2, \dots, 915 \quad (1)$$

Using the fitted model, test

$$H_0 : \beta_j = 0 \quad vs \quad H_1 : \beta_j \neq 0$$

for each predictor. Identify the statistically significant predictors at the 5% level.

(ii) Using the fitted Poisson GLM in Equation (1), the estimated Pearson dispersion parameter is 1.8576. Explain what this value indicates about the relationship between the observed variability and the variance assumed by the Poisson model. What modelling problem does this suggest?

(iii) Using the model specified in Equation (1), fit a quasi-Poisson GLM to the data. Compare the standard errors of the estimated coefficients for **fem** and **ment** with those obtained from the Poisson GLM. Explain why the standard errors from the quasi-Poisson model differ from those of the Poisson model.

(iv) Suppose that a researcher is concerned that the Poisson model specified in Equation (1) does not adequately account for the variability in the article counts. Fit a Negative Binomial model using the same response variable and predictors as in Equation (1), and compare its coefficient estimates and statistical conclusions with those from the Poisson model.

(v) Use the `stepAIC()` function in R to perform stepwise model selection for the Negative Binomial GLM fitted in Part (iv). Compare the final selected model with the full Negative Binomial model from Part (iv) as two nested models. Use the `anova()` function in R to perform a likelihood-ratio/deviance test and determine whether the two nested models differ significantly in their fit.